

How the numbers are made

The logic behind every trip, penalty and plan figure — in six simple pages, with real numbers from our own road.

1 · The big picture

2 · One trip

3 · Queues

4 · Road penalties

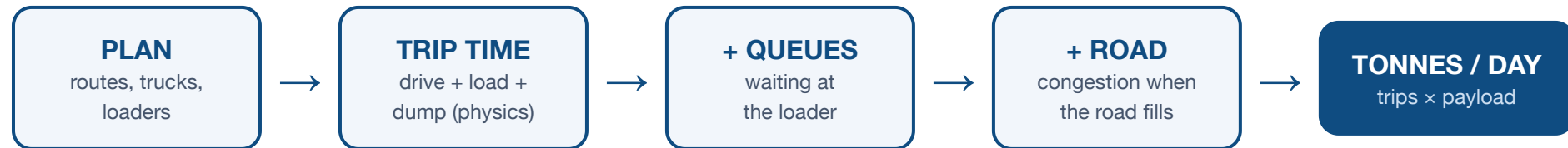
5 · Building a plan

6 · Is it right?

2026-08-26 · prepared from the production model, all values live

From a plan to tonnes: four steps, one clock

Everything the app reports comes from this one pipeline. Every step is calibrated against our own dispatch and GPS history (Jan–Jun 2026).



The one rule behind everything

A truck can only be in one place at a time. Its day is 24 hours; every minute spent driving, queuing or being loaded comes out of that budget.

The model counts those minutes honestly, then:

$$\text{trips per truck per day} = \text{day minutes} \div \text{minutes per trip}$$

Where the numbers come from

- **Distances** — surveyed chainage (same table as the map)
- **Speeds** — official speed-limit sheets + GPS measurements
- **Load / dump times** — calibrated per route from dispatch data
- **Payloads** — measured tonnes per trip per route
- **Traffic shape** — 34 days of GPS, hour by hour, per road

A trip is minutes, and we count every one

Real example: TF → HUAFEI (long haul), 100 trucks, 6 loaders — the model's actual numbers.

Drive loaded + empty
378 min

Load
5 + 1 min

Dump
2 min

Breaks, fuel, dispatch
218 min

+

Drive time = distance ÷ speed

Surveyed kilometres over each section, at the official limit for that section (or the GPS-measured speed where we have it). Loaded and empty directions are priced separately — empty runs faster.

TF→HUAFEI: ~68 km each way → 378 min of driving per round trip.

Overheads are measured, not guessed

The 218 min covers shift breaks, fuelling, dispatch gaps and the shift changeover — calibrated so the model's trips/day matches what each route historically achieves. The red sliver is bunching behind slower trucks (+3 min).

Total ≈ 607 min per trip → 2.4 trips per truck per day.
× 48 t payload × 100 trucks ≈ 11,300 t/day on this route.

More trucks ≠ more tonnes, once the loader is busy

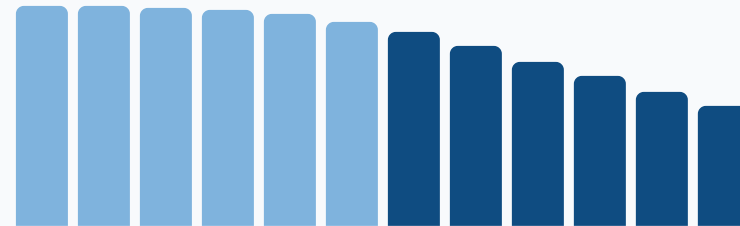
The fleet is a closed loop: N trucks circulate between the road and a few loading faces. We use the standard closed-fleet queue model (machine-repair / M/M/c//N) — the same one used across the mining literature.

How it works

- Each loader serves one truck every ~5 minutes.
- If trucks return faster than the loaders can serve, a queue forms — and queue minutes are added to every trip.
- With too many trucks, output **flattens** at the loaders' maximum: adding trucks then adds only waiting, never tonnes.

Match check: every plan row also respects $\text{trips/hr} \leq \text{loaders} \times (60 \div \text{load minutes})$ — output can never exceed what the loading faces can physically serve.

Trips per truck vs fleet size (one route)



Left: few trucks — everyone loads immediately. Dark bars: the queue grows; each truck completes fewer trips. The **knee** is where adding a truck stops paying for itself, and the app marks it on every saturation chart.

The road has a capacity, and traffic comes in waves

One loaded lane per direction, 50 m following distance. Capacity = the slowest section speed ÷ 50 m — about 400–600 trucks/hour depending on the section.

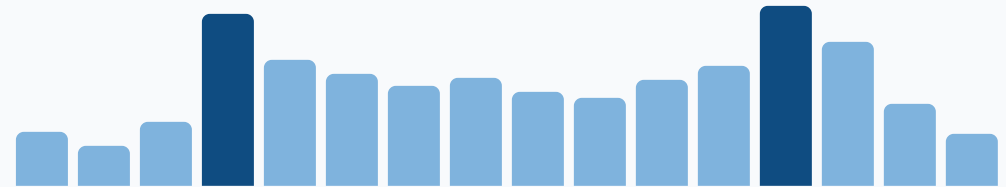
Below capacity → small penalty, above → queue

Up to capacity we use the standard road-engineering curve (BPR): delay grows gently, then sharply as v/c approaches 1. **Past capacity we switch to a queue formula** (Akçelik): delay then grows with *how long* the overload lasts — a polynomial can't say that honestly.

Road load (v/c)	Status	Effect on trip
under 0.7	GREEN	nearly none
0.7 – 1.0	YELLOW	minutes added
over 1.0	RED	queue grows all shift

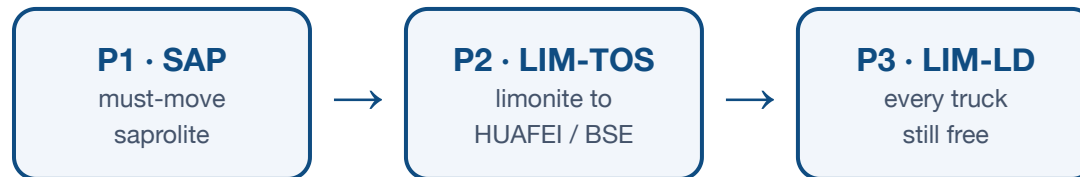
Traffic is waves, not a flat line

From 34 days of GPS: each road has its own rhythm — KR peaks at the 07:00/19:00 shift changes, the coast road in the evening. Hours range from 0.13× to 2.45× the daily average. Since congestion cost rises steeply with load, pricing at the flat average **understates** the real day. We price each hour at its measured level.



measured trucks on road by hour — dark = shift-change peaks

Priorities first, then the leftovers — under the site rules



The standing rules (always enforced)

- **SAP routing:** each pit sends ~2,000 t/day to its POS buffer; the **rest goes direct to FeNi** per the mine plan.
- **BLB accepts RIM trucks only.**
- **POS is transit:** whatever enters must leave on IWIP reclaim trucks (input = output), sized by the same trip model.
- **Targets = the sales table** (SAP 5.72 Mt · LIM-TOS 4.64 Mt · LIM-LD 6.64 Mt for Sep–Dec).
- **Other tenants** (MHM, POSITION, PMA...) take road space, never our tonnage — they make our trips slower, not fewer.

Every row also gets...

- **Weighbridges:** trips split across the eligible bridges (owner matrix), balancing load — the sheet shows each bridge's share, its utilisation %, and its expected waiting time.
- **Loaders:** from the measured trucks-per-loader ratio.
- **Road check:** the whole plan's trucks (plus tenants) are put on the corridor hour by hour, and every section gets a GREEN / YELLOW / RED verdict per hour.

Result: tonnes per route per month, fleet, trips, congestion — all from one pipeline, so Excel, the app and the API can never disagree.

Checked against history, every day, automatically

Backtest against real dispatch data

The model is replayed over **2,124 measured route-days** (Jan–Jun): for each day we predict trips per truck and compare to what actually happened.

~15%

typical error
(best routes 12–13%)

$R^2 > 0.7$

variance explained
on the backtest

=

on par with published
state of the art

A 2026 literature review (115 papers) confirmed: our architecture (physics + closed-fleet queue + road pricing) is the approach the mining research itself recommends for planning tools.

80+ automated gates on every change

- Fleet is conserved — the plan can never invent or lose trucks
- Excel = app, cell by cell, for all saved scenarios
- Congestion advisory never changes tonnage figures
- Contractor walls (BLB=RIM), POS input=output, sales targets
- Model curves frozen and diffed — silent drift is impossible

Principle: every number the app shows states its own basis — measured, calibrated, or rule — so a wrong assumption can be found and corrected, not hidden.